



Tool Order : O.1072613.71-160-3-1-FDP, Partially Shipped, PETROBRAS BRASIL, DEEPWATER ORION, 8-MRL-248H-RJS

Header

GENERAL INFORMATION

Client: PETROBRAS BRASIL
Rig: DEEPWATER ORION
Well: 8-MRL-248H-RJS
Version no.: 5C-10T-1D

SERVICE QUALITY GROUP

FSM DD: IBalderrama **Sales rep:**
FSM CMarculino **Drill**
MWD: **Engg:**
SDM **COR:**
Cell
Name:
Created By: LCarvalho10 on 09 Jan 2026 08:45 AM
Updated By: JSampaio3 on 23 Jan 2026 06:18 PM

SHIPPING INFORMATION

TCO	Estimated	Shipping	Ship TBC	Truck
Priority:		Instructions:		Company:
Dock	PACU	Dock		County:
Name:		Phone:		
Planned	21 Jan 2026	Field:		Country/
Ship Date:				Region:
State/		Basin/		Well
Province:		Project:		Latitude/
				Longitude:

WELL PARAMETERS

Hole Size:	406.4 (mm)	BHA To	true	Bore Hole:	3
		Go:			
Bit Type:	PDC	Sea	(m)	Start	931 (m)
		Depth:		Depth	
				MD:	
End Depth	1760 (m)	End Depth	1760 (m)	Inclination:	0 (Degree)
MD:		TVD:			
Azimuth:	(Degree)	Min	136.274 (m3/hr)	Max	249.837 (m3/hr)
		Planned		Planned	
		Flow Rate:		Flow Rate:	
Min BH	281.95 (K)	Max BH	293.15 (K)	Max BH	298.15 (K)
Circ		Circ		Static	
Temp:		Temp:		Temp:	
Max	(K)	Max BH	272.251 (ATM)	Max	(ATM)
Surface		Pressure:		StandPipe	
Temp:				Pressure:	
Max DLS:	0 (deg/30m)	Mud Type:	Sea Water	Mud	8.549 (lb/gal)
				Density:	
Mud	0 (CP)	Mud PV:	0 (CP)	Mud PV	(K)
Viscosity:				Temp:	
High	false	High	false	High Mud	false
Temperature:		Pressure:		Density:	
High	false	High	false	Corrosive	false
Dogleg:		Shock:		Environment:	
Max		Maxwell		HSPM	
Magnetic		Version:		Version:	
Spacing:					

ToolScope Version:	RX 5/6 Version:	Connect BHA (Heisenberg) Version:
Connect BHA Version:		

INSPECTION PARAMETERS					
Inspection Company:	Drilco	Monitoring Company:	Not Required	Inspection Level:	DS-1 Cat 3-5 Edition 4
Inspection Paperwork:	YES				

ADDITIONAL INFORMATION			
Comments:	8-MRL-248H- RJS_BHA#02_16.5in_PDC+PDM+ARC+ 8-MRL-248H- RJS_BHA#02a_16.5in_PDC+ROT+ARC+SonicScope+MWD_Rev1_20260108_2026.01.14.11.10.202 8-MRL-248H- RJS_BHA#02a_16.5in_PDC+ROT+ARC+SonicScope+MWD_Rev1_20260108_2026.01.14.11.10.040	Basket true required:	
	Hydraulics still pending configuration		

Tools					
		Status	Redirect	Transport Type	Assigned Toolset
O.1072613.71-160-3-1-FDP		Partially Shipped			
Batteries					
Field Replaceable Batteries					
Quantity : 1 (ARC6/8 - ADN6/8 -Sonic6/8 Battery)					
ARC6/8 - ADN6/8 -Sonic6/8 Battery		Released		EQ- T6017257 SN31365	
Additional Tool Specific Instructions (PLEASE READ)	N/A	Battery Type		Regular Temp - NEW (T6017257)	
Battery Status	New	comments		N/A	
FMP Comment	N/A				
Submitted by LCarvalho10 on 09-Jan-2026 08:45					
Quantity : 1 (ARC6/8 - ADN6/8 -Sonic6/8 Battery)					
ARC6/8 - ADN6/8 -Sonic6/8 Battery		Released		EQ- T6017257 SN31366	
Additional Tool Specific Instructions (PLEASE READ)	N/A	Battery Type		Regular Temp - NEW (T6017257)	
Battery Status	New	comments		N/A	
FMP Comment	N/A				
Submitted by LCarvalho10 on 09-Jan-2026 08:45					
DT Equipment					
Hydra AP Drilling Jars					
Quantity : 1 (DJR-DRLG-HYDRA-0950)					
DJR-DRLG-HYDRA-0950		Redirected	BRMEA2-DTSS	Ship To Job	
Application	Standard environment – Regular Make Up Torque Required		Latch Reqd	NO	
Lift Subs Reqd	NO		Stand Off Reqd	NO	

Third Party Insp Reqd	NO	Additional Tool Specific Instructions (PLEASE READ)	9 1/2" Hydra-Jar AP. Firing up and down.
DH Connection Type	PIN	UH Connection Type	BOX
DH Connection Size	7 5/8 REG	UH Connection Size	7 5/8 REG
JAR_Activation	Up and Down	comments	N/A
FMP Comment	under maintenance		

Submitted by CMarculino on 14-Jan-2026 16:53

Quantity : 1 (DJR-DRLG-HYDRA-0950)

DJR-DRLG-HYDRA-0950	Redirected	BRMEA2-DTSS	Ship To Job
Application	Standard environment – Regular Make Up Torque Required	Latch Reqd	NO
Lift Subs Reqd	NO	Stand Off Reqd	NO
Third Party Insp Reqd	NO	Additional Tool Specific Instructions (PLEASE READ)	9 1/2" Hydra-Jar AP. Firing up and down.
DH Connection Type	PIN	UH Connection Type	BOX
DH Connection Size	7 5/8 REG	UH Connection Size	7 5/8 REG
JAR_Activation	Up and Down	comments	N/A
FMP Comment	N/A		

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Drilling Tools and Services

Specialized Tools - WCD-WBPE

Quantity : 1 (MudGard 950)

MudGard 950	Redirected	BRMEA2-DTSS	Ship To Job
DH Connection Size	7 5/8 REG	DH Connection Type	PIN
UH Connection Size	7 5/8 REG	UH Connection Type	BOX
Additional Tool Specific Instructions (PLEASE READ)	9" Mud Gard Filter Sub	comments	N/A
FMP Comment	Mudgard - 5000349622-1 entregue em Pull 24/1/26		

Submitted by CMarculino on 14-Jan-2026 16:53

Quantity : 1 (Well Defender 950)

Well Defender 950	Redirected	BRMEA2-DTSS	Ship To Job
DH Connection Size	7 5/8 REG	DH Connection Type	PIN
Ball Catcher	YES	Ball Catcher Trapper	YES
Ball Catcher Capacity	18	Ball Bypass Required	YES
UH Connection Size	7 5/8 REG	UH Connection Type	BOX
Additional Tool Specific Instructions (PLEASE READ)	Well Commander	comments	N/A
FMP Comment	WD 9,5 - MNB58153-04 WD 9,5 25/1 EM PULL 24 BBCMnb60626-02 - MONTAR 25/1 EM PULL 24		

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Quantity : 1 (Well Defender 950)

Well Defender 950	Redirected	BRMEA2-DTSS	Ship To Job
DH Connection			

Size	7 5/8 REG	DH Connection Type	PIN
Ball Catcher	YES	Ball Catcher Trapper	YES
Ball Catcher Capacity	18	Ball Bypass Required	YES
UH Connection Size	7 5/8 REG	UH Connection Type	BOX
Additional Tool Specific Instructions (PLEASE READ)	Well Commander	comments	N/A
FMP Comment	N/A		
Submitted by CMarculino on 14-Jan-2026 16:53			

Stabilizers			
Quantity : 2 (ILS 900)			
ILS 900 #1	Submitted		
ILS 900 #2	Submitted		
DH Connection Size	7 5/8 H90	DH Connection Type	PIN
UH Connection Size	7 5/8 H90	UH Connection Type	PIN
Stabilizer OD	16 3/8	Additional Tool Specific Instructions (PLEASE READ)	ILS 16 3/8" to be sent connected and torqued DH of SonicScope PRY and BKP
comments	N/A	FMP Comment [ILS 900 #1]	N/A
FMP Comment [ILS 900 #2]	N/A		
Submitted by LCarvalho10 on 09-Jan-2026 08:45			

Subs			
Quantity : 1 (Ball Catcher Sub)			
Ball Catcher Sub	Redirected	BRMEA2-DTSS	Ship To Job
Collar ID	129.54 (mm)	DH Connection Size	7 5/8 REG
DH Connection Type	PIN	Material	Steel
Nominal Length	N/A	Tool OD	9 1/2
UH Connection Size	7 5/8 REG	UH Connection Type	BOX
Additional Tool Specific Instructions (PLEASE READ)	Ball catcher - Sent connected and torqued with well defender.	comments	N/A
FMP Comment	WD 9,5 - MNB58153-04 WD 9,5 25/1 EM PULL 24 BBCMnb60626-02 - MONTAR 25/1 EM PULL 24		
Submitted by CMarculino on 14-Jan-2026 16:53			

Quantity : 1 (Ball Catcher Sub)			
Ball Catcher Sub	Redirected	BRMEA2-DTSS	Ship To Job
Collar ID	129.54 (mm)	DH Connection Size	7 5/8 REG
DH Connection Type	PIN	Material	Steel
Nominal Length	N/A	Tool OD	9 1/2
UH Connection Size	7 5/8 REG	UH Connection Type	BOX
Additional Tool Specific Instructions (PLEASE READ)	Ball catcher - Sent connected and torqued with well defender.	comments	N/A
FMP Comment	N/A		
Submitted by CMarculino on 14-Jan-2026 16:53			

Quantity : 1 (MWD Crossover Subs)			
MWD Crossover Subs		Accepted	MSSC-HC KSBD15624
DH Connection Size	7 5/8 H90	DH Connection Type	PIN
Length	0.5 (m)	Length Tolerance	No Length Sensitive
Material	Monel	Sub OD	228.6 (mm)
UH Connection Size	7 5/8 REG	UH Connection Type	BOX
Vendor	SLB	Additional Tool Specific Instructions (PLEASE READ)	Extra XO for PUP UH standalone
comments	N/A	FMP Comment	N/A
Submitted by LCarvalho10 on 09-Jan-2026 08:45			

Logging While Drilling

Resistivity

Quantity : 1 (ARC 900)			
ARC 900		Accepted	
BHA To Go Tool	true	Pressure Rating	0 - 20000 (PSI)
Battery Loaded Tool	YES	Battery Handling Tool Kit	YES
APWD Service Required	YES	AIM Service Required	NO
Extenders	Uphole Only	High Temperature	false
High Pressure	false	LWD Batteries	YES
Software Version	9.9a84	DH Connection Type	PIN
DH Connection Size	7 5/8 REG	UH Connection Type	BOX
UH Connection Size	7 5/8 H90	Additional Tool Specific Instructions (PLEASE READ)	arcVISION 900
BR2GO_Primary configuration	N/A	BR2GO_Standard Configuration	N/A
BR2GO_DownLink Enabled	N/A	Name	
FMP Comment	N/A	comments	N/A
Submitted by LCarvalho10 on 09-Jan-2026 08:45			

Sonic

Quantity : 1 (Sonic 900)			
Sonic 900		Accepted	SD9C-AA G1975
UH Connection Type	BOX	Additional Tool Specific Instructions (PLEASE READ)	sonicVISION with ILS 16 3/8" connected DH.
Software Version	7.1b-00	DH Connection Size	7 5/8 H90
DH Connection Type	PIN	Extenders	Downhole only
LWD Batteries	NO	UH Connection Size	7 5/8 REG
comments	N/A	FMP Comment	N/A
Submitted by CMarculino on 22-Jan-2026 10:27			

Quantity : 1 (SonicScope 900)			
SonicScope 900		Accepted	
Software Version	8.7	DH Connection Size	7 5/8 H90
DH Connection Type	PIN	Extenders	Downhole only
Lower Stabilizer OD	10 (slick sleeve)	LWD Batteries	NO
SonicScope			

Support Kit Required	NO	UH Connection Size	7 5/8 REG
UH Connection Type	BOX	Upper Stabilizer OD	10 (slick sleeve)
Additional Tool Specific Instructions (PLEASE READ)	SonicScope 900 with ILS 16 3/8" connected DH.	BHA To Go Tool	true
Basket test required?	true	Fishing Diagrams required?	true
Battery loading required?	false	Bit size (in)	419.099 (mm)
SCLP (mud slowness)	N/A	Mud Density	8.55 (lb/gal)
AcqStopDelay	N/A	Downlink Enabled	N/A
CFG_Main	N/A	CFG_Slow	N/A
CFG_Medium	N/A	CFG_Fast	N/A
CFG_Salt	N/A	CFG_CE RM	N/A
CFG_PO	N/A	CFG_ADV	N/A
comments	N/A	FMP Comment	N/A

Submitted by LCarvalho10 on 09-Jan-2026 08:45

Measurements While Drilling

Mud Pulse

Quantity : 1 (Rhossili 900)

Rhossili 900 **Accepted** **MDCIX-HA|E2920**

DH Connection Size	7 5/8 H90	UH Connection Size	7 5/8 H90
DH Connection Type	PIN	UH Connection Type	BOX
Additional DH Sub for String Connection	NO	Additional UH Sub for String Connection	NO
Stab Saver Sub required	NO	Stabilizer Type	Not Required
Stabilizer Size	N/A	Stabilizer Made Up	NO
Agitator / Agitator Plus (BHA Oscillation System)	NO	High Telemetry Frequency	NO
Zero Gap MMA	NO	Silicate Mud	NO
Special Mud (non-Silicate)	NO	MOC #	N/A
Rotary Switch Enabled	NO	If TruLink, is DDS service required?	YES
Additional Tool Specific Instructions (PLEASE READ)	Trulink 900 slick. Flow kit 600-1200gpm. Gap = 0.16	Planned Min Flow Rate	136.26 (m3/hr)
Planned Max Flow Rate	Min MW 8.5ppg at 1100gpm - tool signal 136.5 psi	Mud Density	8.55 (lb/gal)
MMA Rotor Type	249.81 (m3/hr)	MMA Gap	0.16 (in)
MTA Flow Range	5.9 Standard (300-1400GPM)	MWD WOB / TOR Services	YES
MWD PWD	600 - 1200 (GPM)		D&I - 4 Axis Shock - TruLink Drilling Dynamics
High Temperature	NO	Software Version	16_0_b07
Telemetry Mode	false	Total G	N/A
Total H	QPSK 10.5 Hz / 6 bps	Dip Angle	-43 (Degree)
Delayed Survey	N/A	Survey on CRPM Cessasion	NO
TF Switch	NO	Extenders	Uphole & Downhole
DH Filter Sub	Low Inclination (2.5 < incl. < 3.5)	Collar Type	Non-ROP Collar preferred
BHA To Go Tool	YES	Reamer/hole opener above theTeleScope	NO
Reamer/hole opener fluid loss	true	BR2GO_Send survey after (min)	N/A
BR2GO_Enable	0 (%)	BR2GO_Send utility frame after	N/A
CRPM at start	N/A		
BR2GO_BHA			

Power sharing	N/A	BR2GO_No CRPM check	N/A
BR2GO_Health MODF	N/A	BR2GO_DDS interval	N/A
BR2GO_Auto survey mode	N/A	BR2GO_FSL	N/A
BR2GO_Downlink bit time period (sec)	N/A	BR2GO_TS Marker recording	N/A
BR2GO_Latitude	N/A	BR2GO_Collar RPM	N/A
BR2GO_Magnetic declination	N/A	BR2GO_Collar Type	N/A
BR2GO_Primary configuration	N/A	BR2GO_Health MODF from run beginning	N/A
comments	N/A	FMP Comment	N/A
Submitted by CMarculino on 22-Jan-2026 15:46			
Quantity : 1 (TeleScope 900)			
TeleScope 900		Submitted	
BHA To Go Tool	true	Reamer/hole opener above theTeleScope	NO
Reamer/hole opener fluid loss	0 (%)	Planned Min Flow Rate	136.26 (m3/hr)
Planned Max Flow Rate	249.81 (m3/hr)	Mud Density	8.55 (lb/gal)
MMA Rotor Type	5.9 Standard (300-1400GPM)	MMA Gap	0.16 (in)
MTA Flow Range	600 - 1200 (GPM)	MWD WOB / TOR	YES
MWD PWD	NO	Services	D&I - 4 Axis Shock
High Temperature	false	PMEA modified for 180W capability	NO
Software Version	V14.0C04	Telemetry Mode	QPSK 10.5 Hz / 6 bps
Total G	N/A	Total H	N/A
Dip Angle	-43 (Degree)	Delayed Survey	NO
Survey on CRPM	NO	TF Switch	Low Inclination (2.5 < incl. < 3.5)
Cessasion			
Extenders	Uphole & Downhole	DH Filter Sub	NO
Collar Type	Non-ROP Collar preferred	DH Connection Size	7 5/8 H90
UH Connection Size	7 5/8 H90	DH Connection Type	PIN
UH Connection Type	BOX	Additional DH Sub for String Connection	NO
Additional UH Sub for String Connection	NO	Stab Saver Sub required	NO
Stabilizer Type	Not Required	Stabilizer Size	N/A
Stabilizer Made Up	NO	Agitator / Agitator Plus (BHA Oscillation System)	NO
High Telemetry Frequency	NO	Zero Gap MMA	NO
Silicate Mud	NO	Special Mud (non-Silicate)	NO
MOC #	N/A	Rotary Switch Enabled	YES
Additional Tool Specific Instructions (PLEASE READ)	TeleScope 900 slick. Flow kit 600-1200gpm. Gap = TBC	BR2GO_Health MODF	N/A
	Min MW 8.5ppg at 1100gpm - tool signal 136.5 psi		
	Max MW 9.5ppg at 1100gpm - tool signal 151.7 psi		
BR2GO_Magnetic declination	N/A	BR2GO_Send survey after (min)	N/A
BR2GO_Downlink			

bit time period (sec)	N/A	BR2GO_Latitude	N/A
BR2GO_BHA Power sharing	N/A	BR2GO_Health MODF from run beginning	N/A
BR2GO_FSL	N/A	BR2GO_Send utility frame after	N/A
BR2GO_Auto survey mode	N/A	BR2GO_No CRPM check	N/A
BR2GO_Primary configuration	N/A	BR2GO_TS Marker recording	N/A
BR2GO_Enable CRPM at start	N/A	BR2GO_Collar RPM	N/A
comments	N/A	FMP Comment	N/A

Submitted by LCarvalho10 on 09-Jan-2026 08:45

Surface

Slider

Quantity : 1 (Slider)

Slider	Redirected	BRMEA2-DTSS	Ship To Job
Additional Tool Specific Instructions (PLEASE READ) comments	N/A	Additional Tool Specific Instructions (PLEASE READ) FMP Comment	Slider hammer para mudgard. N/A

Submitted by CMarculino on 14-Jan-2026 17:00

Support Equipment

Quantity : 1 (TRANSPORTATION BASKET)

TRANSPORTATION BASKET	Accepted		
comments	N/A	FMP Comment	N/A

Submitted by LCarvalho10 on 09-Jan-2026 08:45